

Scale-It-Up Blueprint

Unit 3: Culinary Academy · Lesson 3-6 · EduWonderLab

UNIT 3 · LESSON 3-6 · TEACHER NOTES

Standard 6.RP.3

FORMAT Scaling design challenge

TIME 25–35 min

GROUPING Pairs

TOPIC Use Ratio Reasoning

STANDARD 6.RP.3

PREP LEVEL Light — print blueprint cards, scaling table, graph paper

Learning goal *I can use ratio reasoning to scale a design and keep every pair proportional.*

Teacher setup

- Print one blueprint set per pair. Each card scales a recipe, map, or design.
- Students fill in missing values and check that every pair stays proportional.
- Include “too small/too large” distractor answers for partner critique.

Materials

Blueprint cards, Scaling table, Graph paper, Pencil.

Differentiation & language support

- Scaling table organizes original and new amounts side by side (SPED).
- Sentence stem: “I scaled by ___, so ___ becomes ___.”
- ESOL: use real contexts — recipes, maps, models.

Key vocabulary (English / Español):

Term	Español	What it means
scale	escalar	to make larger or smaller by the same factor
proportional	proporcional	keeping the same ratio

Quick teacher check: *Ask a pair to show the scale factor used on Card 5.*

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Name: _____ Date: _____ Class: _____

Your goal: *I can use ratio reasoning to scale a design and keep every pair proportional.*

What you need

Blueprint cards, Scaling table, Graph paper.

How to play

1. Read the original amounts on the card.
2. Find the scale factor the card asks for.
3. Scale every value by the same factor.
4. Check that the new ratio matches the original.

Blueprint Cards

Use ratio reasoning to scale each design. Keep every pair of values proportional.

1. A recipe for 4 people uses 2 cups rice. For 12 people?	2. Model scale 1 in : 24 in. A 5-in model car is how long in real life?
3. Map scale 1 in : 50 mi. How far is 4 in on the map?	4. 3 cups flour makes 12 cookies. Flour for 36 cookies?
5. 2 gallons of paint cover 700 sq ft. Coverage for 6 gallons?	6. Blueprint 1 cm : 2 m. A wall is 8 cm on the plan. Real length?
7. A dog eats 5 lb of food in 2 weeks. How much in 6 weeks?	8. Class ratio 3 boys : 5 girls. If there are 18 boys, how many girls?

Scaling table (original → scaled):

Explain your thinking

Explain how you can check that your scaled design stayed proportional.

Sentence frame: *I scaled by ____, so every amount was multiplied by the same number.*

★ **Challenge:** *Card 8 scaled by 6. If 2 more boys joined to make 20 boys, would the ratio still be exactly 3 : 5? Explain.*

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. A recipe for 4 people uses 2 cups rice. For 12 people? Answer: 6 cups ($\times 3$)	2. Model scale 1 in : 24 in. A 5-in model car is how long in real life? Answer: 120 in ($\times 24$)
3. Map scale 1 in : 50 mi. How far is 4 in on the map? Answer: 200 mi	4. 3 cups flour makes 12 cookies. Flour for 36 cookies? Answer: 9 cups ($\times 3$)
5. 2 gallons of paint cover 700 sq ft. Coverage for 6 gallons? Answer: 2,100 sq ft ($\times 3$)	6. Blueprint 1 cm : 2 m. A wall is 8 cm on the plan. Real length? Answer: 16 m
7. A dog eats 5 lb of food in 2 weeks. How much in 6 weeks? Answer: 15 lb ($\times 3$)	8. Class ratio 3 boys : 5 girls. If there are 18 boys, how many girls? Answer: 30 girls ($\times 6$)

Teaching notes & look-fors

- Every value must be multiplied by the same factor to stay proportional.
- Challenge: 20 boys is not a multiple of 3, so 3 : 5 would not hold exactly.